

PROJECT EXPO 2026

Smart IV Drip Monitoring Using IOT

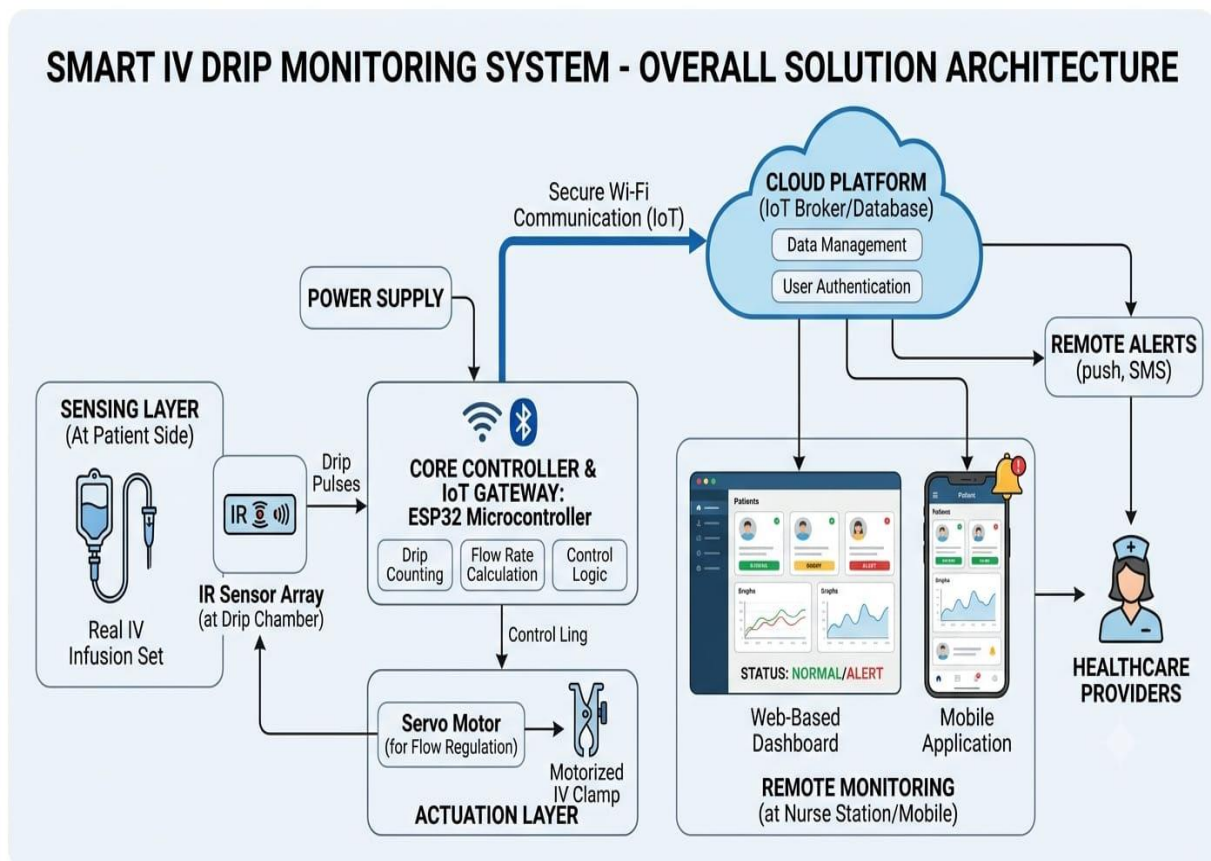
TEAM MEMBERS:

M.DHASLIMA BANU

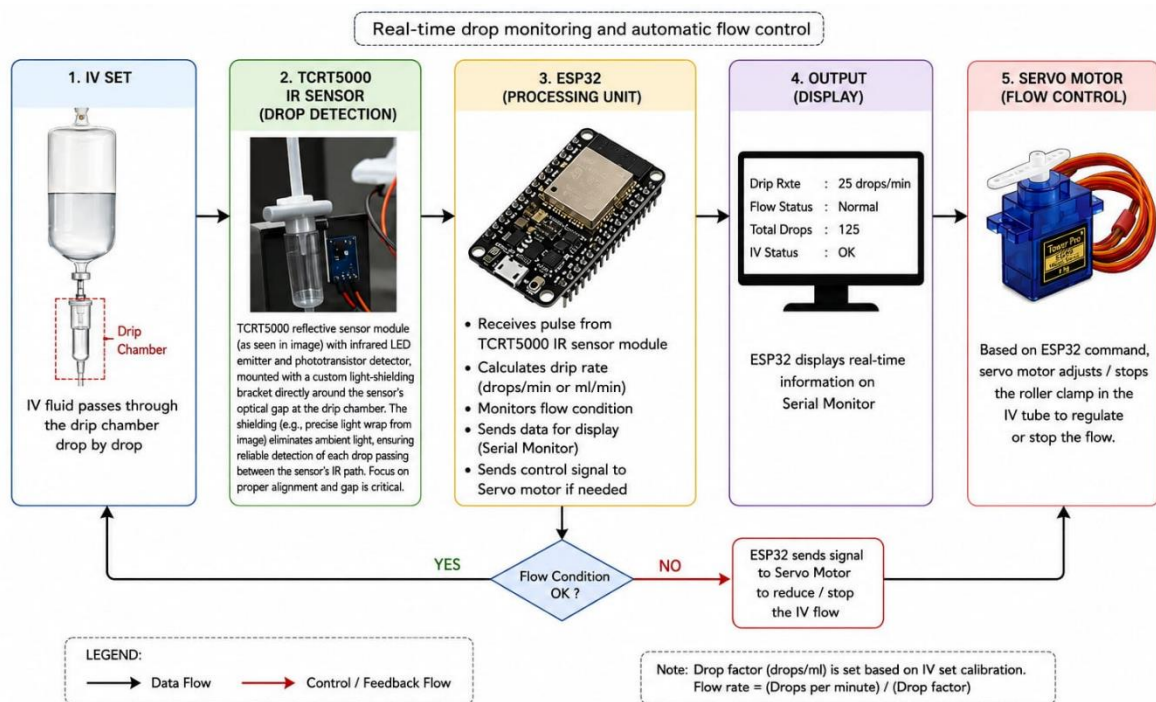
P. GANGASHREE

SYSTEM ARCHITECTURE:

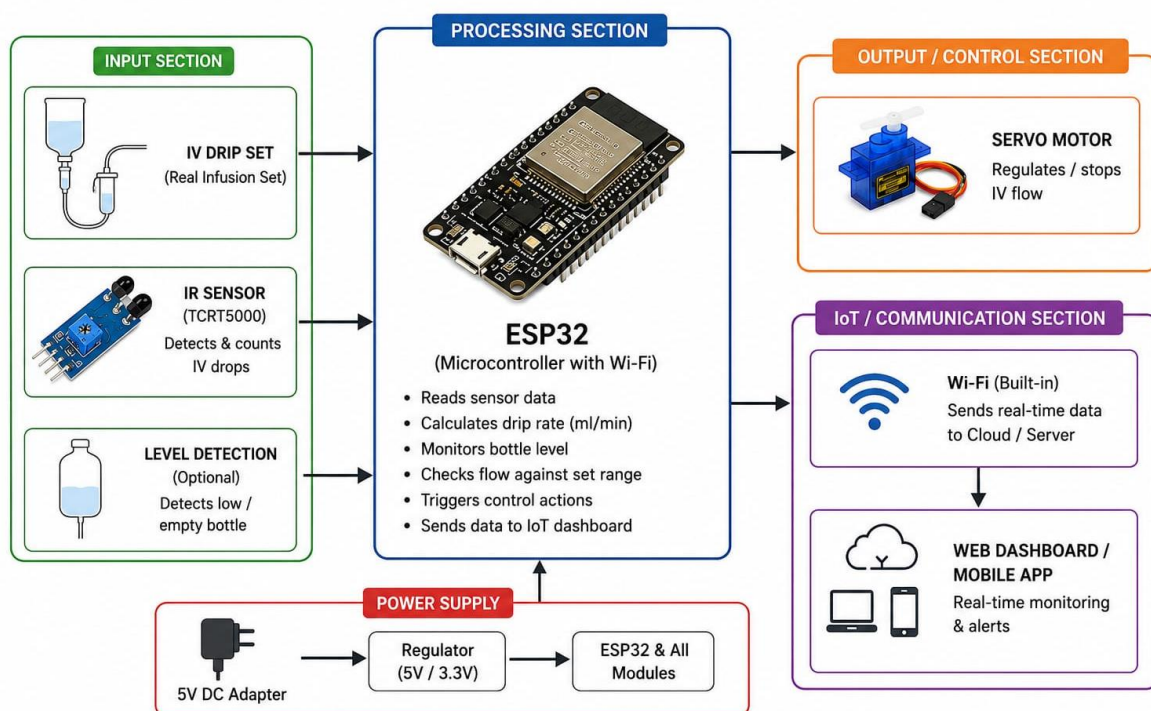
OVERALL SOLUTION ARCHITECTURE:



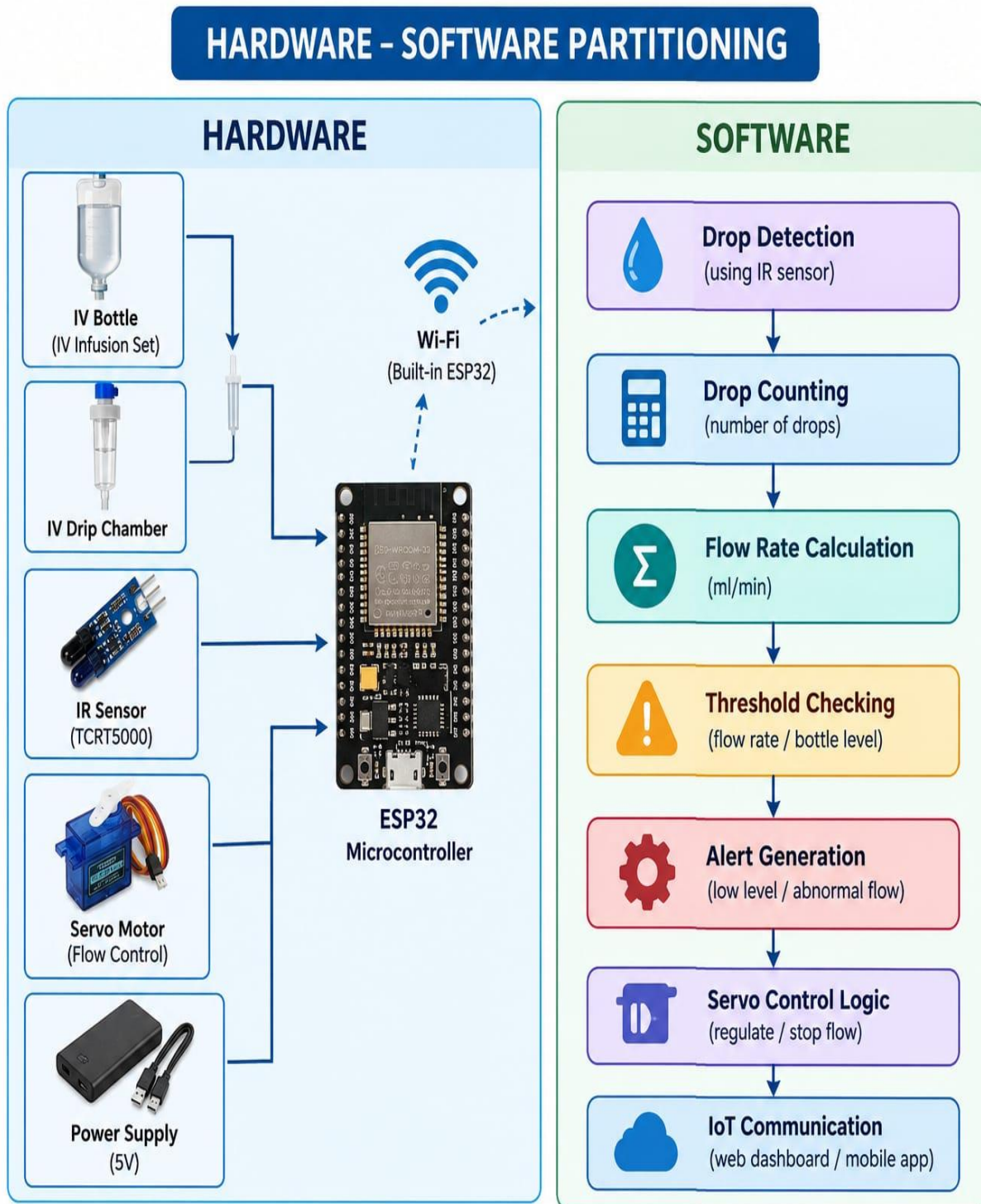
DATA FLOW DIAGRAM:



FUNCTIONAL BLOCK DIAGRAM:

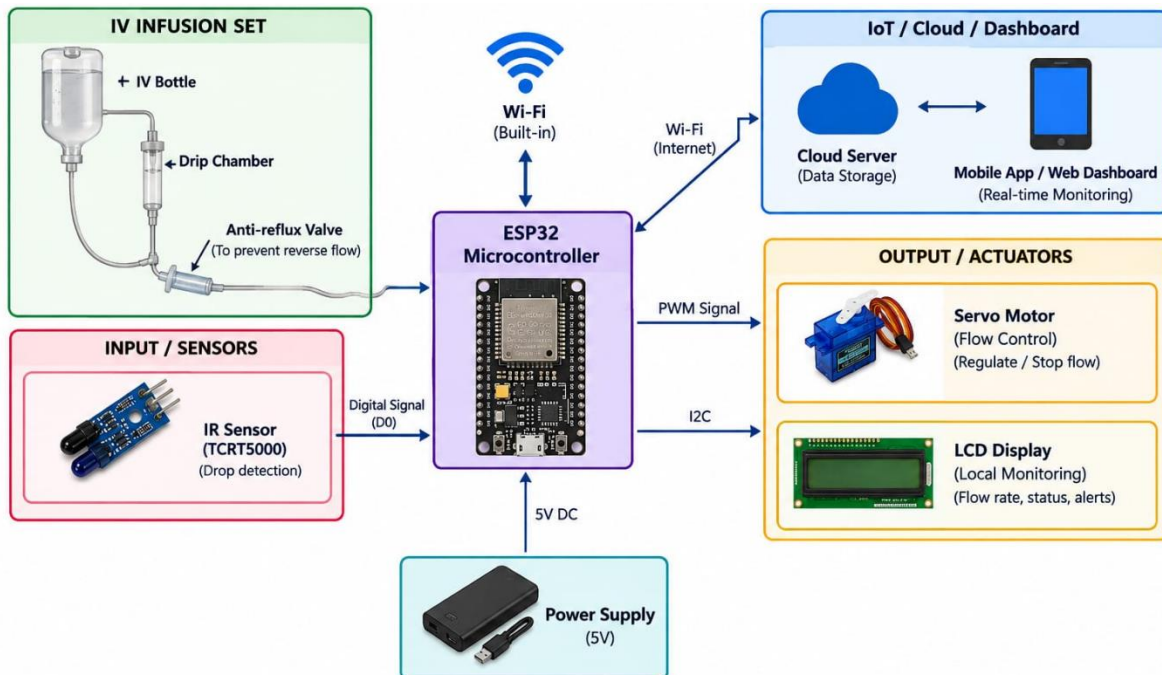


HARDWARE-SOFTWARE PARTITIONING:

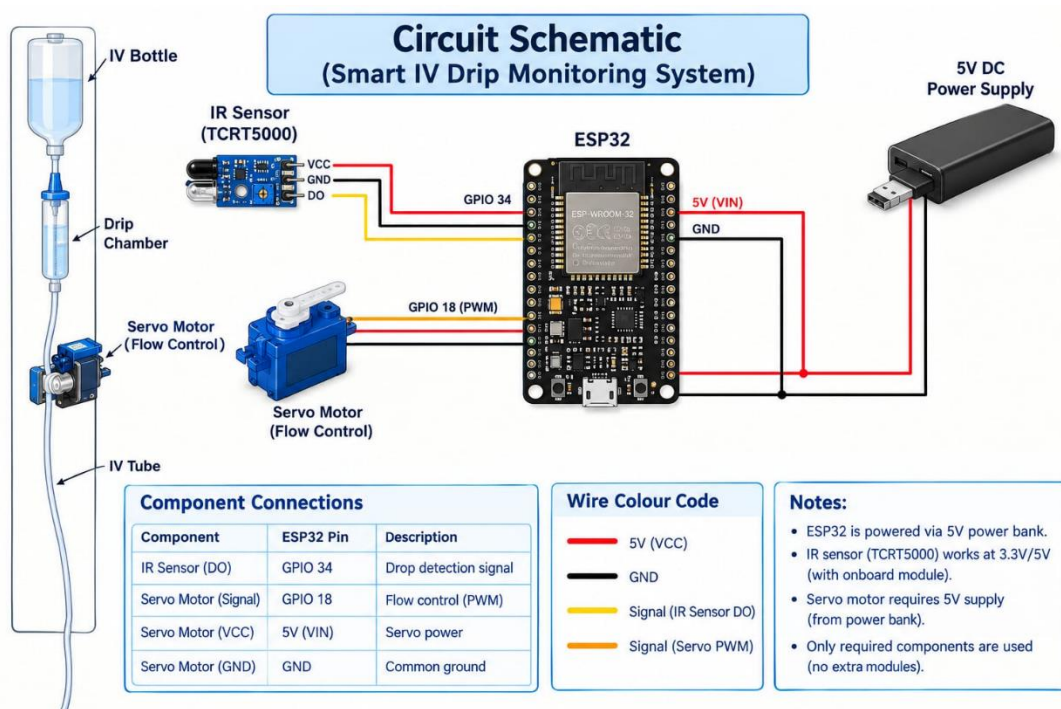


HARDWARE DESIGN:

FINAL BLOCK DIAGRAM:



CIRCUIT SCHEMATICS:



COMPONENTS SELECTION:

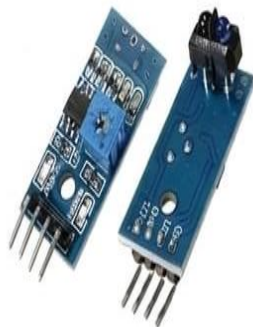
SMART IV DRIP MONITORING SYSTEM USING ESP32

1 ESP32 DEVELOPMENT BOARD



Main controller of the system. Provides Wi-Fi for IoT communication and processes sensor data.

2 TCRT5000 IR SENSOR MODULE



Detects and counts the drops passing through the drip chamber using IR light interruption.

3 SG90 SERVO MOTOR



Controls the IV tube roller clamp to regulate or stop the liquid flow based on ESP32 commands.

4 JUMPER WIRES



Used for making electrical connections between components on the breadboard.

4 IV SET WITH DRIP CHAMBER & BOTTLE



Provides the complete liquid path and container. The drip chamber and sensor are used for drop detection.

5 POWER SUPPLY



Provides regulated 5V power to the ESP32 board and servo motor.

MINIMUM REQUIRED COMPONENTS

- ✓ ESP32 Development Board
- ✓ TCRT5000 IR Sensor Module
- ✓ SG90 Servo Motor
- ✓ IV Set with Drip Chamber & Bottle
- ✓ Jumper Wires & 5V Power Supply

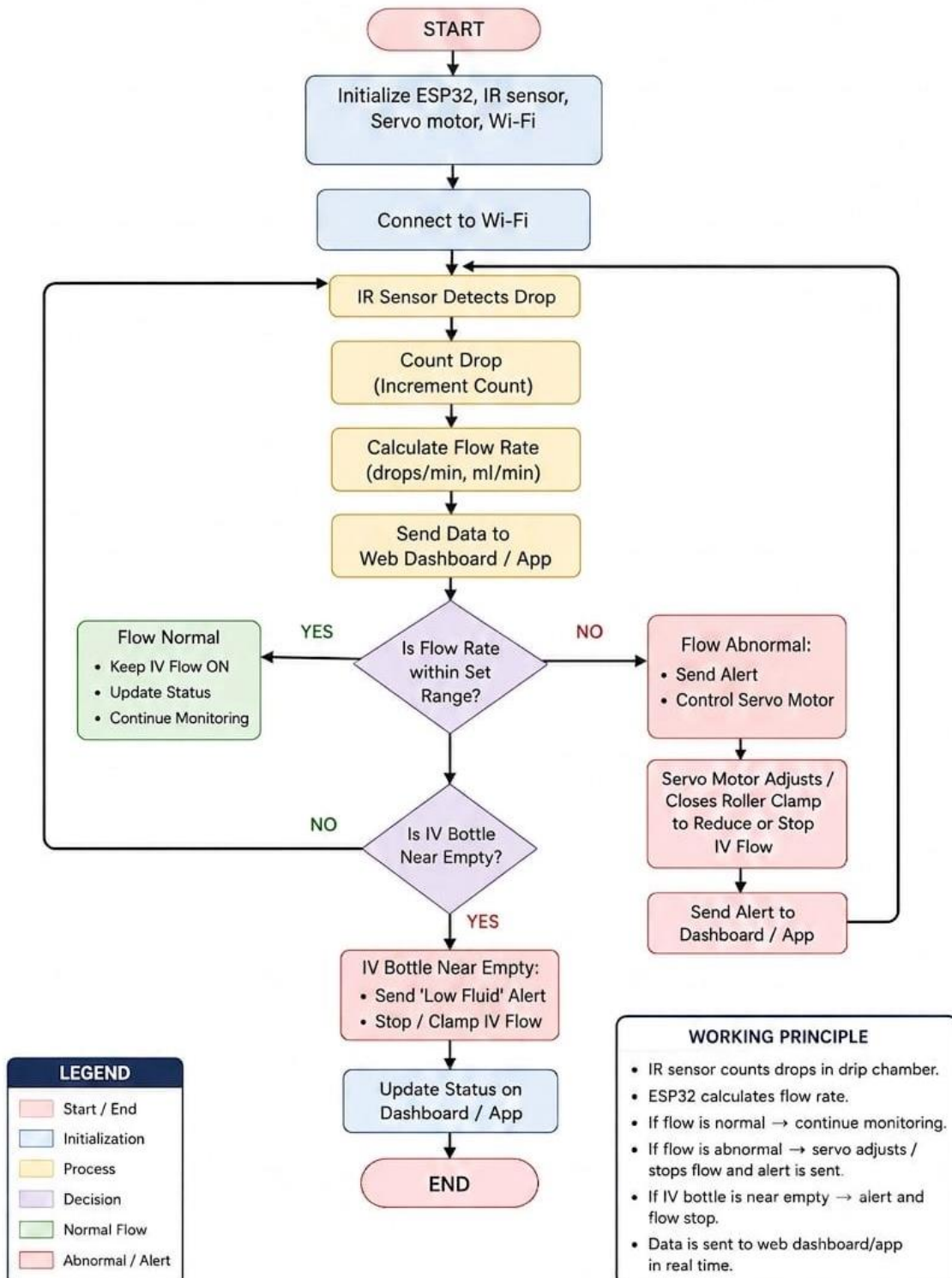
NOTE
LCD Display, Buzzer, and LEDs are optional additions for enhanced local UI and alerts.

BILL OF MATERIAL(BOM):

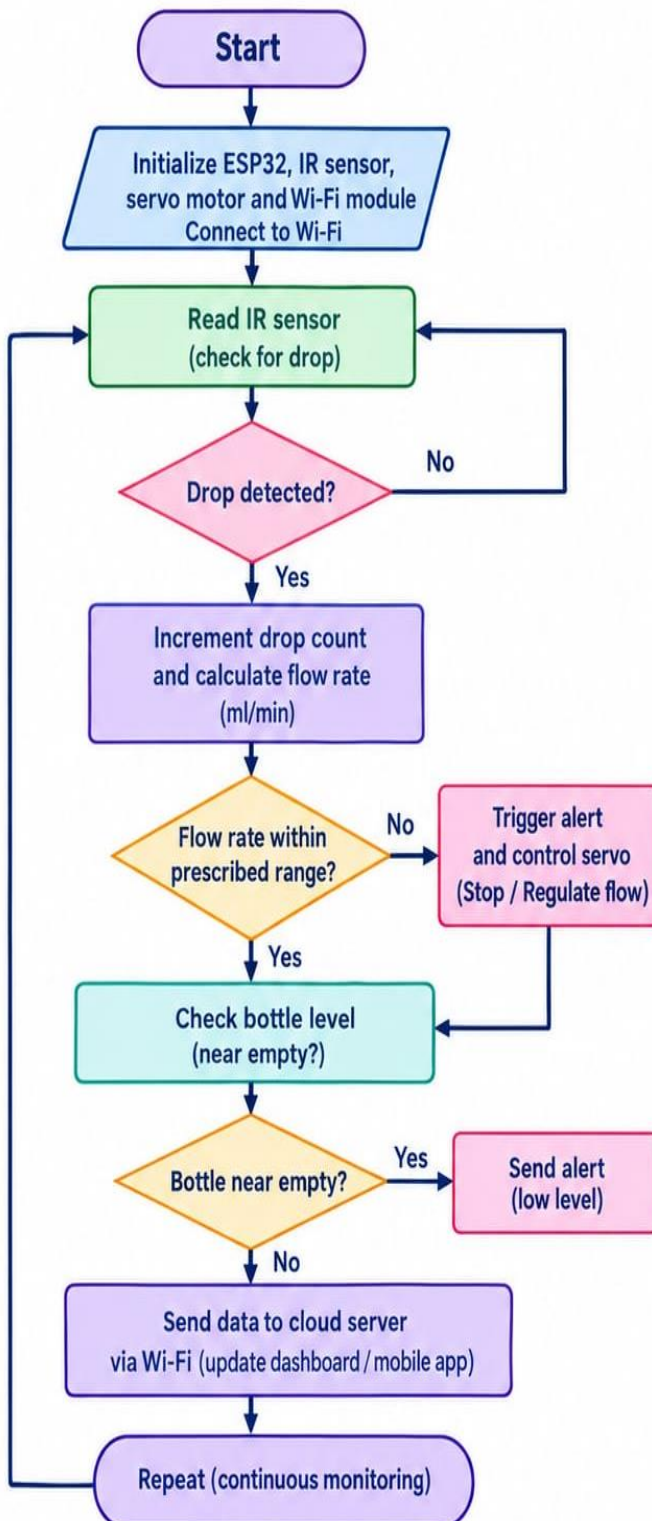
S.No.	Component Name	Specification / Model	Quantity	Unit Price (₹)	Total (₹)
1	ESP32 Wi-Fi Module 	ESP32 WROOM, 40 Pins	1	375	375
2	Servo Motor 	SG90	1	98	98
3	IR Sensor (Line Tracking Module) 	TCRT5000 (Line Tracking Module)	1	39	39
4	Jumper Wires 	—	1 set	15	15
5	IV Infusion Set 	Standard IV Set	1	180	180
				TOTAL COST	₹707

SOFTWARE DESIGN:

FLOWCHART:



ALGORITHM:



Algorithm Steps

- 1 Initialize ESP32, IR sensor, servo motor and Wi-Fi module. Connect to Wi-Fi.
- 2 Read IR sensor to detect a drop in the drip chamber.
- 3 If a drop is detected, increment the drop count and calculate the flow rate (ml/min).
- 4 Check if the flow rate is within the prescribed range. If not, trigger an alert and control the servo motor to stop/regulate the flow.
- 5 Check the bottle level (near empty). If yes, send a low bottle alert.
- 6 Send the data to the cloud server via Wi-Fi (update web dashboard / mobile app).
- 7 Repeat the process continuously for real-time monitoring.

Key Variables

Drop Count	Number of drops detected
Flow Rate	ml/min
Prescribed Flow Rate	Set value (as per doctor's order)
Bottle Level	% or remaining volume

COMMUNICATION PROTOCOLS:

